# Self Learning: Will You Carry the Umbrella? (Conditional Entropy)

Date: 25.07.2025 (Friday)

Duration: 1 Hour

## Scenario

We observe Weather (W) and whether people carry an Umbrella (U). From several days of data, the joint probability distribution is:

| | U = Yes | U = No |
|---|---|---|
| Rainy | 0.5 | 0.1 |
| Sunny | 0.2 | 0.2 |

• Weather (W): {Rainy, Sunny}

• Umbrella (U): {Yes, No}

## Questions

1. What is the marginal probability that the weather is Rainy?

2. What is the marginal probability that a person carries an umbrella?

3. What is the most likely pair (Weather, Umbrella) to occur?

4. What does the conditional entropy H(W|U) represent in this context?

5. Which of the following represents the correct formula for Conditional Entropy H(W|U)?

   a) H(W|U) = H(W,U) - H(U)
   b) H(W|U) = H(U,W) - H(W)
   c) H(W|U) = ∑u P(u) H(W|U=u)
   d) All of the above

6. Compute the conditional entropy H(W|U).

7. Compute the conditional entropy H(U|W).

8. If someone is not carrying an umbrella, what is the most likely weather?

9. Compare H(W|U) and H(U|W). Which one is larger, and what does this tell us about the relationship between Weather and Umbrella?

10. Verify the chain rule of entropy: H(W,U) = H(W|U) + H(U).

11. Self-Learning: If the umbrella decision was completely independent of the weather, how would H(W|U) and H(U|W) change? Explain and prove your reasoning.

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